

Injury Severity Prediction for Road Casualties in Great Britain: Comparing Linear and Ensemble Classifiers on STATS19 Administrative Data

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Abstract—This study investigates whether administrative collision-record features from the 2023 Great Britain STATS19 road safety dataset can predict severe road casualty outcomes. Severe casualties are defined as fatal or serious injuries, while slight injuries form the non-severe class. The collision, vehicle, and casualty tables were merged at casualty level, producing 132,977 records and 42 selected raw predictors. All train/test and cross-validation splits were grouped by collision identifier to reduce leakage. Logistic regression, Random Forest, and XGBoost were compared using macro-F1 as the primary metric. The final XGBoost model achieved macro-F1 = 0.619, Severe recall = 0.649, and ROC-AUC = 0.739 on the held-out test set. Grouped SHAP analysis identified casualty type, vehicle manoeuvre, and vehicle type as the strongest predictors. A sensitivity experiment showed that random oversampling did not improve Severe-class detection over cost-sensitive XGBoost

Keywords— STATS19, road safety, injury severity, XGBoost, Random Forest, grouped cross-validation, SHAP, imbalanced classification.

I. INTRODUCTION

A. Dataset and Research Context

Road traffic collisions remain a public health and road-safety burden in Great Britain. In 2023, the Department for Transport recorded 132,977 road casualties of all severities, including 1,624 fatalities and 29,711 killed or seriously injured casualties [1]. Predicting injury severity from administrative collision records can support post-collision risk auditing, road safety planning, and identification of high-risk road-user groups.

This study uses the 2023 Great Britain STATS19 road safety dataset, which consists of three linked administrative tables: collisions, vehicles, and casualties, covering road environment, vehicle characteristics, and individual casualty attributes, respectively. Machine learning has increasingly been applied to crash-severity prediction, with ensemble models such as Random Forest and XGBoost often performing strongly on structured administrative data [2, 3]. However, many studies focus mainly on collision-level predictors, while casualty-level and vehicle-level characteristics may provide additional information about the injured person and their crash context. In addition, row-wise validation can allow casualties from the same collision to appear in both training and evaluation sets, creating collision-level leakage

The final modelling dataset was produced by merging all three tables at casualty level, yielding 132,977 casualty records and 42 selected raw features spanning casualty characteristics,

vehicle and driver attributes, road infrastructure, environmental conditions, temporal indicators, and collision-context aggregates. The binary target variable is Severe, defined as Fatal or Serious injury (20.8% of records), versus Slight (79.2%). Fatal casualties represented only 1.2% of records, making a three-class formulation unreliable and justifying the binary collapse into a single Severe class, Fig 1.

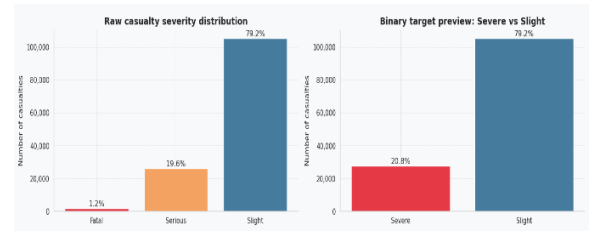


Figure 1- Target Distribution

B. Research Questions

The main research question is: Can administrative collision-record features predict Severe casualty outcomes in Great Britain, and which factors drive these predictions? A Severe outcome is defined as Fatal or Serious injury, while Slight injury is treated as the non-severe class. Three sub-questions guide the analysis:

RQ1: Do tuned tree-based ensemble models, specifically Random Forest and XGBoost, outperform a regularised Logistic Regression baseline on imbalanced STATS19 data?

RQ2: Which administrative feature groups most strongly drive Severe-outcome predictions, as identified through grouped SHAP analysis?

RQ3: Does random oversampling of the minority Severe class improve Severe-class detection compared with cost-sensitive class weighting?

C. Exploratory Analysis

Severe rates varied substantially across features: single-vehicle collisions showed a 30.1% Severe rate versus 19.3% for multi-vehicle collisions, and Severe rate increased with speed limit, reaching approximately 28% at 60 mph. These patterns motivated the multi-level feature design. These patterns motivated the multi-level feature design and justified combining Fatal and Serious into a single Severe class.

II. RELATED WORKS

Road casualty severity prediction has increasingly used machine learning methods, particularly ensemble classifiers and

imbalance-handling techniques. Tong [2] applies Random Forest, XGBoost, SMOTE, and SHAP to 2023 STATS19 pedestrian injury data, highlighting the importance of road and collision-context features. However, that study focuses mainly on pedestrian-involved collisions and collision-level predictors.

Le [3] proposes a safety-oriented LightGBM framework for binary KSI prediction on UK STATS19 data from 2020 to 2024, incorporating SMOTE, threshold optimisation, probability calibration, and SHAP. The present work differs by comparing three model families under a shared grouped validation protocol and applying a safety-aware threshold rule that preserves Severe-class recall.

Broader crash-severity studies show that class imbalance must be handled explicitly. Boo and Choi [4] compare Random Forest, Extra-Trees, and XGBoost with resampling strategies, while Amiri et al. [5] compare SMOTE, ADASYN, random oversampling, and threshold moving for pedestrian crash severity, finding that cost-sensitive and threshold-based strategies can outperform resampling for minority-class detection. For interpretability, this study uses SHAP [6] and TreeExplainer [7] to explain the final XGBoost model after model selection. These findings support the decision to evaluate cost-sensitive weighting and random oversampling separately rather than treating either as a default solution

III. METHODOLOGY

A. Data Preparation and Target Definition

The modelling unit was one casualty record. The STATS19 casualty table was merged with the vehicle table using `collision_index` and `vehicle_reference`, then collision-level variables were added using `collision_index`. Key uniqueness checks confirmed no duplicate rows were introduced during merging.

The binary target was derived from `casualty_severity`. Fatal and Serious casualties were coded as `Severe = 1`, while Slight casualties were coded as `Severe = 0`. This binary formulation was chosen because Fatal casualties represented only 1.2% of the data, making a separate Fatal class too sparse for reliable modelling. The grouping also follows killed-or-seriously-injured road-safety framing used in related STATS19 work [3]. Severity-related variables were removed before modelling to prevent target leakage.

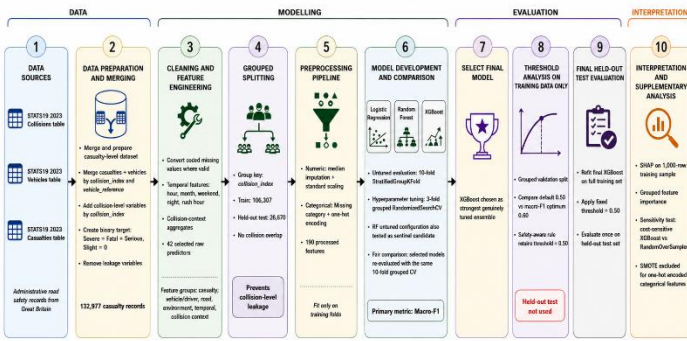


Figure 2. Modelling pipeline:

STATS19 uses coded values such as -1, 9, and 99 to represent missing, unknown, or not-applicable values in some variables. These were converted to missing values only where confirmed by the official coding guide, because the same numeric code can have different meanings across variables. Missing values were not imputed before splitting; instead, imputation was placed inside the modelling pipeline to prevent validation or test data from influencing preprocessing

B. Feature Engineering

The final feature set covered six groups: casualty characteristics, own-vehicle and driver features, road infrastructure, environmental conditions, temporal features, and collision-context aggregates. Temporal variables were derived from date and time fields because severity risk can vary by hour, night-time driving, and weekend travel. Collision-context aggregates summarised all vehicles involved in the same collision, allowing each casualty record to include both their own vehicle information and the wider crash environment.

C. Splitting, Preprocessing, and Leakage Prevention

All train/test and cross-validation splits used `collision_index` as the grouping key, because multiple casualties from the same collision share collision-level features such as road type, speed limit, weather, and vehicle-context aggregates. Grouping by `collision_index` prevented collision-level leakage throughout all splits and folds. The grouped 80/20 split produced separate training and held-out test sets with no collision overlap.

Preprocessing used a `ColumnTransformer` [11]. Numeric features were median-imputed and standardised. Median imputation was used because age and vehicle-age variables were skewed, making the median more robust than the mean. Categorical STATS19-coded variables were imputed with a dedicated Missing category and one-hot encoded. Keeping preprocessing inside the pipeline assured the fact that imputation, scaling, and encoding were fitted only on training folds during cross-validation.

D. Models and Hyperparameter Tuning

Three classifiers were compared against a majority-class dummy baseline. Logistic Regression with L2 regularisation and balanced class weights was used as a clear linear baseline. Random Forest was included because bagged decision trees can identify non-linear feature association patterns and are widely used for tabular classification [8]. XGBoost was included because gradient-boosted trees are effective for structured tabular data and have been used in crash-severity prediction studies [2, 9]. Class imbalance was handled using balanced class weights for Logistic Regression and Random Forest, and fold-specific `scale_pos_weight` for XGBoost.

Hyperparameter tuning used 3-fold grouped RandomizedSearchCV [10] with macro-F1 as the refit metric. Random search was chosen because it efficiently explores large hyperparameter spaces with a constrained evaluation budget [10]. Search spaces covered regularisation strength for Logistic Regression, tree structure and sampling parameters for Random Forest, depth, learning rate, and subsampling parameters for XGBoost.

The tuning budget was constrained for computational feasibility. Random Forest's untuned configuration was also evaluated as a sentinel candidate to ensure a strong baseline was not excluded by the limited search. All tuned configurations were re-evaluated under the same 10-fold grouped protocol as the untuned baselines for a fair comparison.

E. Evaluation Metrics, Threshold Selection, and Interpretation

Macro-F1 was the primary evaluation metric because the target was imbalanced and macro-F1 gives equal weight to both classes. Additional metrics included G-Mean, balanced accuracy, Severe precision, Severe recall, Severe F1, average precision, and ROC-AUC to capture both class-balanced performance and the Severe-class precision-recall trade-off.

Threshold selection used a grouped validation split from the training data only, so the held-out test set remained untouched until final evaluation. The threshold was chosen using a safety-aware rule: maximise macro-F1 while preserving default-threshold Severe recall. This was appropriate because missing Severe casualties is undesirable in a road-safety context.

SHAP values were computed after final model fitting on a 1,000-row training sample, not on the held-out test set. This ensured that interpretation did not influence model selection or test performance. A supplementary sensitivity analysis compared cost-sensitive XGBoost with RandomOverSampler to test whether duplicating minority-class Severe cases improved detection. SMOTE was excluded because interpolation on one-hot encoded categorical features can create invalid fractional indicators [12].

IV. RESULTS

A. Data Pre-processing

The three STATS19 tables merged to 132,977 casualty records with 42 selected raw features, expanding to 190 processed features after preprocessing. The target was imbalanced: 79.2% Slight and 20.8% Severe, with Fatal casualties representing only 1.2%. junction_control had the largest remaining missingness at 45.2%, handled inside the preprocessing pipeline.

B. Model Comparison

Table 1 reports mean metrics across the fair 10-fold grouped cross-validation comparison. Tuning improved XGBoost most clearly, increasing macro-F1 from 0.588 to 0.623. Wilcoxon signed-rank tests on fold-level macro-F1 provided exploratory evidence that XGBoost tuning improved consistently across all 10 grouped folds ($p = 0.002$), while Logistic Regression tuning was negligible ($p = 0.557$) and Random Forest showed identical fold scores before and after tuning ($p = 1.000$). These tests are treated as exploratory because cross-validation folds are not fully independent. Random Forest achieved the highest overall macro-F1, but XGBoost was selected as the final model because it was the strongest genuinely tuned ensemble and outperformed tuned Logistic Regression under the same grouped protocol. This supports RQ1, with the qualification that Random Forest's strong result came from the unchanged sentinel baseline rather than successful tuning.

Table 1. 10-fold grouped CV results

Model	Version	Macro-F1	G-mean	Sev-recall	ROC-AUC
Random Forest	Untuned/Tuned	0.632	0.669	0.608	0.747
XGBoost	Tuned*	0.623	0.674	0.649	0.743
Logistic Regression	Tuned	0.602	0.672	0.703	0.739
XGBoost	Untuned	0.588	0.666	0.731	0.738
Dummy	Baseline	0.442	0.000	0.000	0.500

*Final model.

Random Forest tuned = untuned: sentinel retained.

C. Threshold Selection and Final Test

The pure macro-F1 threshold was 0.60 but raising the threshold from 0.50 to 0.60 reduced validation Severe recall from 0.625 to 0.467. The default threshold of 0.50 was therefore retained under the safety-aware rule. Table 2 reports final held-out test performance.

Metric	Value	Metric	Value
Macro-F1	0.619	Severe recall	0.649
G-Mean	0.67	Severe precision	0.361
Balanced accuracy	0.67	Severe F1	0.464
ROC-AUC	0.739	Average precision	0.43

Table 2. Final held-out test results: XGBoost, threshold = 0.50

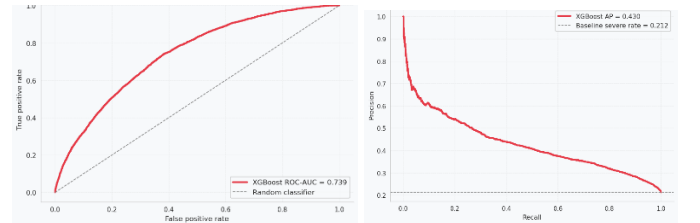


Figure 3. ROC curve (left, AUC = 0.739) and precision-recall curve (right, AP = 0.430) for the final XGBoost model on the held-out test set. Dashed lines show the random classifier baseline (ROC) and Severe-class prevalence rate of 0.212 (PR curve).

The model correctly identified 3,673 of 5,658 Severe casualties. The predicted severe rate was 38.1% against an actual Severe rate of 21.2%, showing that the model is recall-oriented. The average precision score of 0.430 is above the Severe-class baseline rate of 0.212, indicating useful ranking ability for Severe-casualty detection.

D. Model Interpretation

Grouped SHAP identified casualty_type (3.87), vehicle_manoeuvre (2.64), and vehicle_type (2.16) as the three strongest original STATS19 predictors. These results address RQ2: the model relies mainly on casualty role/type, vehicle behaviour, road context, environmental conditions, and crash dynamics. This supports the value of the three-table casualty-level merge. These features are plausible predictors because

they directly describe road-user vulnerability and crash dynamics rather than only road environment conditions. The dominance of `casualty_type` is consistent with road-safety evidence that vulnerable road users such as pedestrians and motorcyclists face substantially higher severe-injury risk [1].

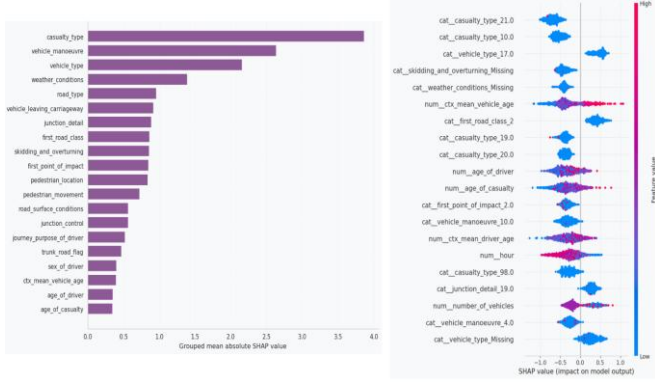


Figure 4. SHAP interpretation of the final XGBoost model.

Notably, these casualty and vehicle-level predictors would be entirely unavailable in collision-level-only analyses such as Tong [2] and Le [3], confirming that the three-table merge adds predictive information beyond road environment features alone.

E. Oversampling Sensitivity

RandomOverSampler produced a negligible macro-F1 gain of 0.001 while reducing Severe recall by 0.018, from 0.649 to 0.632. G-Mean decreased from 0.670 to 0.666, average precision from 0.430 to 0.428, and ROC-AUC from 0.739 to 0.737. Cost-sensitive weighting using `scale_pos_weight` was therefore the more effective strategy for this dataset, addressing RQ3.

V. DISCUSSION

A. Model Performance and Selection

The final XGBoost model achieved held-out macro-F1 of 0.619 and ROC-AUC of 0.739, showing that STATS19 administrative features contain useful but imperfect signal for binary casualty-severity prediction. This is broadly consistent with recent STATS19 studies reporting similar ranking performance for tree-based models [2, 3]. Tree-based ensembles outperformed Logistic Regression on macro-F1, partially supporting RQ1. Random Forest achieved the highest overall macro-F1, but its selected configuration was identical to the untuned sentinel. Since the RF search evaluated only 15 of 64 possible combinations, this should be interpreted as evidence that constrained tuning did not improve RF, not that RF could not be improved further. XGBoost showed the clearest genuine improvement after tuning and was therefore selected as the final model. Its held-out performance reflects a safety-oriented precision-recall trade-off: the model identified 64.9% of Severe casualties, but only 36.1% of casualties predicted as Severe were truly Severe. This means the model is more suitable as a post-collision risk-screening tool than as a definitive severity classifier.

B. Feature Importance and the Three-Table Merge

Grouped SHAP supports the value of the three-table merge. The strongest feature groups, including `casualty_type`, `vehicle_manoeuvre`, and `vehicle_type`, come from the casualty and vehicle tables and would be unavailable in collision-level-only analyses. These predictors are plausible because they describe road-user vulnerability, vehicle behaviour, road context, and crash dynamics. However, SHAP explains model behaviour rather than causal effects, so feature importance should not be interpreted causally. The importance of `casualty_type` is consistent with road-safety evidence that vulnerable road users face higher severe-injury risk [1].

C. Imbalance Handling and Limitations

The oversampling experiment showed that RandomOverSampler did not improve Severe-class detection over cost-sensitive XGBoost, consistent with evidence that resampling does not always outperform threshold or cost-sensitive approaches for minority-class crash severity prediction [5].

Limitations: The study uses only 2023 data, so temporal generalisation is untested. Probability calibration was not assessed, despite the predicted Severe rate exceeding the actual Severe rate. The tuning budget was constrained by time and computational feasibility, so better Random Forest or XGBoost configurations may exist under a wider search. Future work could explore SMOTENC or oversampling before encoding to test whether synthetic minority-class generation improves detection. Finally, spatial information was limited to road class and urban/rural classification, so geographic aggregation could support more targeted road-safety analysis in future work.

VI. CONCLUSION

This study showed that 2023 STATS19 administrative records can distinguish Severe from Slight casualty outcomes with moderate predictive performance. The final XGBoost model achieved macro-F1 = 0.619, Severe recall = 0.649, and ROC-AUC = 0.739 on the held-out test set. The results show that tree-based ensembles improved macro-F1 over Logistic Regression, although Random Forest's strongest configuration came from the unchanged sentinel baseline rather than successful tuning. XGBoost was selected as the strongest genuinely tuned ensemble.

The main contributions are casualty-level modelling using all three STATS19 tables, grouped validation by collision identifier to reduce leakage, and a structured comparison of Logistic Regression, Random Forest, and XGBoost using imbalance-aware metrics. SHAP identified `casualty_type`, `vehicle_manoeuvre`, and `vehicle_type` as the dominant model predictors, confirming the value of the three-table merge. Random oversampling did not improve Severe-class detection compared with cost-sensitive XGBoost.

Overall, the model is best interpreted as a post-collision risk-screening tool rather than a definitive severity classifier. Future work should extend the analysis to multiple STATS19 years, assess probability calibration, and explore spatial aggregation for geographically targeted road-safety interventions.

REFERENCES

- [1] Department for Transport. (2024). Reported road casualties Great Britain, annual report: 2023. UK Government.
- [2] Tong, Y. (2025). Non-linear determinants of pedestrian injury severity: Evidence from administrative data in Great Britain. arXiv:2512.04022.
- [3] Le, K. G. (2026). Safety-oriented and explainable machine learning for KSI crash risk prediction: Evidence from the United Kingdom. PLOS ONE, 21(4), e0347873.
- [4] Boo, Y., & Choi, Y. (2022). Comparison of mortality prediction models for road traffic accidents: An ensemble technique for imbalanced data. BMC Public Health, 22, 1476.
- [5] Amiri, M. A., Afshari, S., & Soltani, A. (2025). Machine learning approaches to traffic accident severity prediction: Addressing class imbalance. Transportation Research Interdisciplinary Perspectives.
- [6] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765–4774.
- [7] Lundberg, S. M., Erion, G., Chen, H., et al. (2020). From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2, 56–67.
- [8] Breiman, L. (2001). Random forests. Machine Learning, 45, 5–32.
- [9] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794.
- [10] Bergstra, J., & Bengio, Y. (2012). Random search for hyperparameter optimization. Journal of Machine Learning Research, 13, 281–305.
- [11] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- [12] Lemaître, G., Nogueira, F., & Aridas, C. K. (2017). Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning. Journal of Machine Learning Research, 18(17), 1–5.